

Developing a Player and Team Comparison & Analytics Tool for Yale Women's Soccer

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1 Abstract

The Yale Women's Soccer team competes in NCAA Division I and in the Ivy League Conference. The coaching staff would like to make decisions informed by data, but there is no qualified staff member who can turn the data into insights. The data is coming from Wyscout, which is a scouting platform owned by Hudl, where certain event data is available for every subscribed user. The variables available on the player level slightly differ from the variables available on the team level, but in general we have access to basic metrics such as goals, assists, expected goals, types of passes and percentages of success, types of actions and percentages of success, challenges, and many more. Using the data available for all Ivy League teams and players, we built a Shiny app for various plots that can be used to compare any players from the Ivy League, and also analyze and compare Ivy League teams. We use scatter plots and radar charts to effectively compare players (including goalkeepers). Radar plots are also used for team comparison, while bar charts are made to analyze and compare the teams in the Ivy League. Our platform gives an easy and accessible way to the coaching staff to use the data that is available without getting overwhelmed or distracted. Our aim for this platform is to help coaches with data-driven roster building and opponent scouting. The platform can be updated when new data is available. With lack of funding for Wyscout's API access, this updating process could only be done manually, but it is certainly possible to do.

2 Introduction

The Yale Women's Soccer had limited success in the Ivy League in the last four years. The coaching staff lacks personnel with analytics background, and the Yale Athletics Department does not want to invest significant amount of money in either good quality of data or a data analyst. This project aims to fill in this gap a bit by presenting the available data in a way that is more consumable to the coaching staff. A huge inspiration for this project is coming from FACTS Consultancy, a Netherlands-based company creating transfer reports for soccer agencies. Instead of reports, we built a Shiny App, which contains customizable visuals. Our goal is to provide the coaches with an analytical tool that helps them make data-driven roster building decisions, hopefully resulting in more wins and a more successful Ivy League campaign next year.

The data is coming from a scouting platform called Wyscout. We collected data for every Ivy League player and team. Data is available as one row per game for each player/team, from their first collegiate game (earliest season available is 2021) to their last game in 2024. This also means that older players and regular starters have more data than younger players. Table 1 below shows the variables available for outfield players, goalkeepers, and teams respectively.

Section 3 contains data cleaning, which reveals the steps we took to transform the data to a format that is usable for comparison. In Section 4, we build multiple visualizations that serve as a basis for our [Shiny App](#). Section 4.1 discusses the design choices for player plots.

Table 1: Variables available for outfield players, goalkeepers, and teams.

Outfield Players	Goalkeepers	Teams
match	match	date
comp	comp	match
date	date	comp
pos	pos	duration
min	min	team
actions	goals_conc	scheme
actions_succ	xcg	goals
goals	shots_against	xg
ass	saves	shots
shots	saves_reflex	shots_on_target
shots_target	exits	shots_pct
xg	long	passes
pass	long_succ	passes_accurate
pass_succ	shortp	pass_pct
long	shortp_succ	poss_pct
long_succ	gkicks	losses
cross	short_gkick	losses_low
cross_succ	long_gkick	losses_med
drib		losses_high
drib_succ		rec
duels		rec_low
duels_won		rec_med
aerial		rec_high
aerial_won		duels
interc		duels_won
loss		duels_pct
loss_own		
rec		
rec_opp		
yellow		
reds		

In Section 4.2 we show the visuals for goalkeepers, while team plots are discussed in Section 4.3. In Section 5, we report some interesting insights, such as the biggest reason behind Yale’s lack of success. Finally, we discuss conclusions, recommendations and ideas for future work in Section 6, including a proposed data pipeline for next seasons.

3 Data Cleaning

3.1 Outfield Players and Goalkeepers

The data from Wyscout comes in excel sheets where each player has their own excel file and each row is one game in which the player played at least 1 minute. So we built a loop to read in all the excel files, automatically adding the player's name and team to new columns, and identifying the position of the player. Goalkeepers were then added to the goalkeeper data frame, `gk`, while outfield players to the player data frame, `df`.

Next, for both goalkeepers and outfield players, we first removed every row where the given player played less than 15 minutes, then removed all players (with all of their data) who did not have a total of 90 minutes of playing time. This is an important step to make sure we get fair and useful comparison between players. Too little game time can paint a very inaccurate picture of a player. One might play 5 minutes and score 1 goal, but that does not mean this player would score 18 goals had they played 90 minutes. Similarly, if someone does not play at least for 90 minutes in a season, they very likely going to be an outlier. These thresholds of 15 minutes and 90 minutes were set arbitrarily, and one could argue for a different threshold. We believed overall this threshold paints a fair picture of the Ivy League trends of playing time.

The data from Wyscout has very specific position labels, such as LCB (left center back) or RCMF (right center midfielder). Every team has different shapes and positions, so to be able to fairly compare players from different teams, we decided to simplify positions. We mapped each Wyscout position to the big four positional categories: GK, DEF, MID, ATT (`gk` did not need this type of cleaning, the positional labels were correct for goalkeepers). We did this on the game level, so each player might have more than one position associated with them. A player might also play in multiple positions in a given game. In this case, we decided to assign only one position for this player in this game, and we chose this one position to be their most common position based on all of their other games.

Since not all players play 90 minutes every game, we decided to transform the raw variables into 'per 90' statistics. For this, we divided the raw statistic with minutes played, then multiplied it by 90. Per 90 statistics show the values we expect a player would generate had they played 90 minutes. We did this for both `gk` and `df` with their respective variables.

The Wyscout data includes both total and successful actions, but we wanted to look at percentages when comparing players. Therefore, we easily created percentage data by dividing the successful count by the total count for certain metrics. We did this for both `gk` and `df` with their respective variables.

3.2 Teams

Data on the team level looks different. Each team has one excel file per season where each game is represented in two rows. The first row is the team the file belongs to and the second

row is their opponent for the given game. We built a loop again to read in all the excel files adding a column for the season variable automatically. We added each file into a team data frame, `dt`. Since every Ivy League game appears twice, we got rid of the duplicates.

Next, we wanted to create columns for the opponent's statistics. Instead of having these in separate rows, we wanted to move it to the row of the given team. First, we created these new columns and called them `opp.[metric]`, then we filled these in. If a row has an odd index number, we filled in using the data from the row below it, and if a row has an even index number, we filled in using the data from the row above it. After this process, we removed each row that belongs to a non-Ivy League team.

The `match` column in `dt` contains the information about which team played at home and which played at away. The two team names are separated by a dash, and the first team (before the dash) played at home. We split `match` into `home_team` and `away_team`.

4 Visualizations in our Shiny App

Our Shiny App consists of three main tabs (**Player**, **Goalkeeper**, and **Team**) with two sub-tabs each that contain the different visuals. In this section we are going to discuss these visualizations in detail.

4.1 Outfield Players

Outfield players are shown under the **Player** tab with two possible visualizations to choose from: scatter plots and radar charts. In the **Player** tab all Ivy League players are available to analyze, who are above the threshold for minutes played – we set this threshold during our data cleaning process. Players have data from all the games they played in an Ivy League team starting the 2021 season.

Ivy League Soccer Analysis

Player
Goalkeeper
Team

Scatter

Radar

Step 1: Choose one or more players to include in the scatter plot.

Select Players:

A. Bray

A. Bryant

A. Butcher

A. Chang

A. Kirschner

A. Miller

A. Moos

A. Solomon

A. Thorvaldsdottir

A. So

B. Golden

C. Laureano

E. Rappole

K. Hernes

L. Booker

L. Jacobs

M. Akins

M. Phillips

N. Yang

P. Ryan

R. Lundstedt

R. Róth

R. Exley

T. Cahalan

T. Teik

V. Lundstedt

Step 2: Choose the type of scatter plot to display. Each type uses different metrics.

Select Plot Type:

Finishing Efficiency

Step 3: Filter data by position. 'All' shows all data available for each player. A specified position will show data only for that position for each player. Specific positions only available if selected players share a position. Players might have different positions each game, therefore they might have data for more than one position available.

Select Position:

All

Step 4: Choose what to compare selected players against. This controls the background coloring. 'League' shows the league average, 'position' shows the average for the selected position, and 'selected players' shows the average of players on the plot.

Figure 1: Screenshot of the front page of our Shiny App

4.1.1 Scatter Plots

The scatter plot for outfield players is our most advanced visual. The user can choose which player(s) they want to include in the plot. Then, the user can decide which metrics they want to look at. There are ten pre-set plot types.

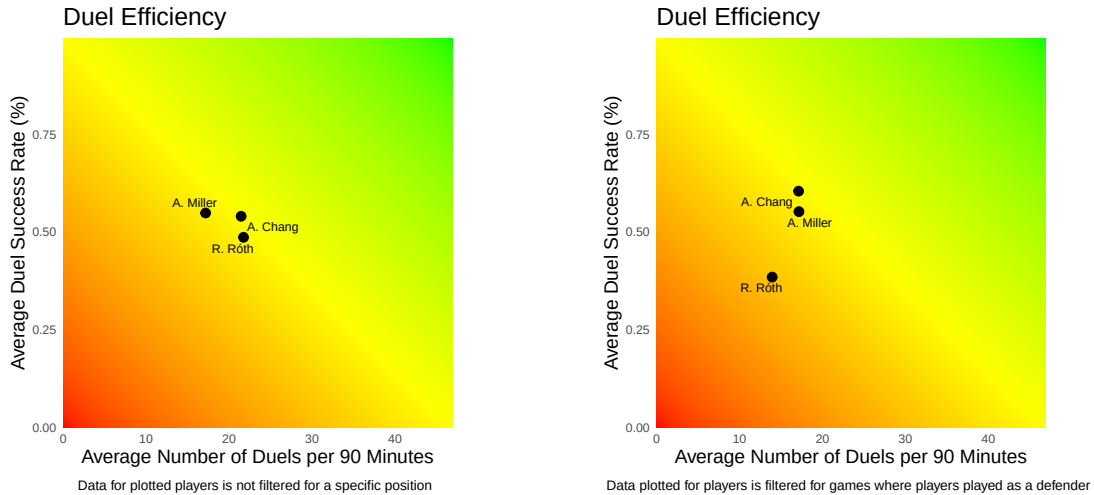
Table 2: Preset Scatter Plot Types and Metrics

Plot Title	X-Axis	Y-Axis
Finishing Efficiency	goals_per90	xg_per90
Long Passing Ability	long_per90	long_pct
Duel Efficiency	duels_per90	duels_pct
Shooting Efficiency	shots_per90	shots_pct
Possession Impact	loss_per90	rec_per90

Table 2: Preset Scatter Plot Types and Metrics (*continued*)

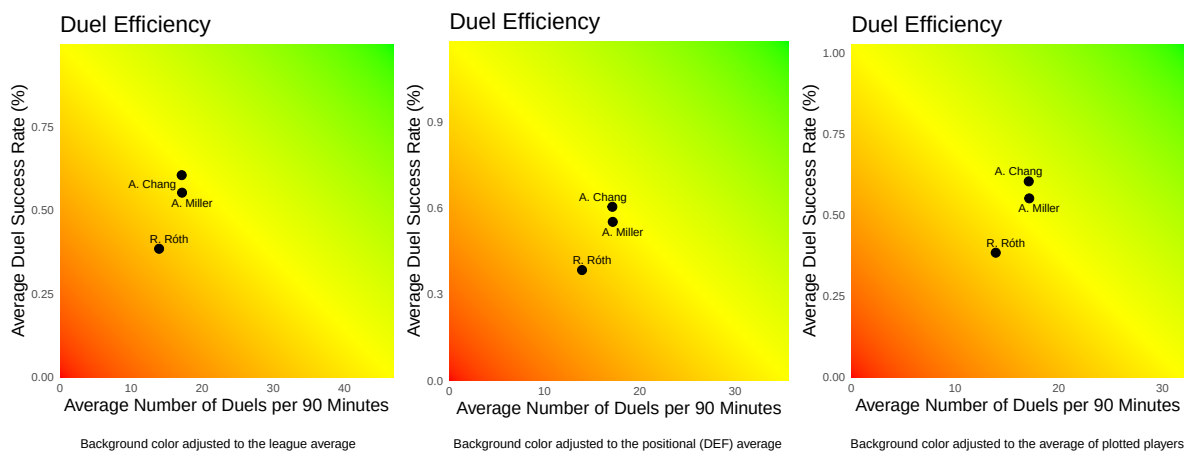
Plot Title	X-Axis	Y-Axis
Aerial Duel Performance	aerial_per90	aerial_pct
Passing Efficiency	pass_per90	pass_pct
Dribbling Effectiveness	drib_per90	drib_pct
Overall Action Efficiency	actions_per90	actions_pct
Crossing Ability	cross_per90	cross_pct

Next, the user picks if they want to restrict the data to a certain position. They can pick any position that all the selected players share (this could be DEF, MID, or ATT). If one of these is selected, the data automatically filters the rows for these players where their position value matches the user-selected position. If the selected players don't all share a single position, the position setting stays at **All**, in which case all data from each player is represented on the plot regardless of position. Selecting a specific position can be extremely useful when the user wants to compare players who often play multiple positions.



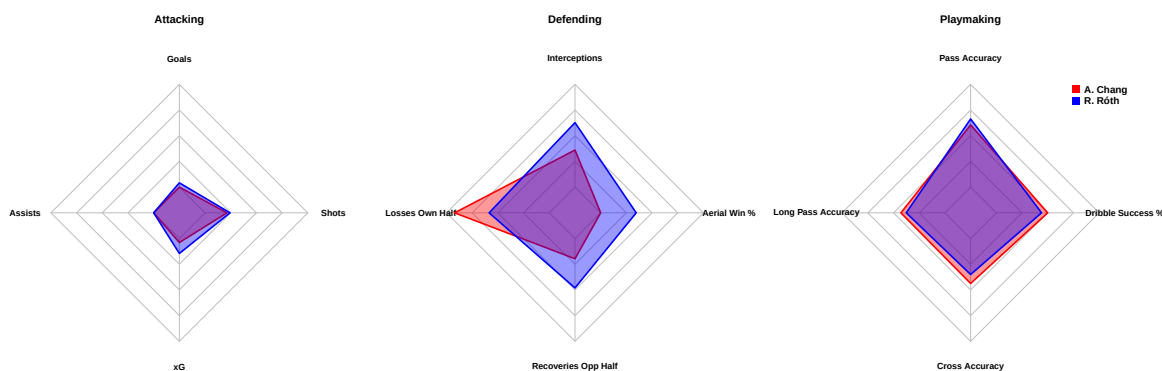
And lastly, the user can control the gradient background on the plot. The gradient is set up so the user can easily tell the quality of players. The color is controlled by the average value, which the user can control. By default it is set to **League**, which means the mid-point of both axes are set to be the league average of the given variable. So in this case, a player in the green area is considered to be a good player in the Ivy League. The other options include **Position** and **Selected Players**. If the **Position** option is chosen, the mid-point of both axes are set to be the average of the given variable among players in the given position. In this case, a player in the green area is considered to be a good player in the Ivy League in the given position. If the **Selected Position** is chosen, the mid-point of both axes are set to be

the average of the given variables among only the players selected to be included in the plot. In this case, a player in the green area is considered to be a good player compared to the other players on the plot.



4.1.2 Radar Charts

Radar chart is a complex and efficient way to compare two players. It has the advantage of including many variables at the same time while still keeping it easy to interpret. We decided to categorize the metrics as **Attacking**, **Defending**, and **Playmaking** skills resulting in three radar charts plotted right next to each other. Our radar charts are percentile based, which means that we standardized all of the metrics to be between 0 and 1. This automatically creates some sort of ranking among all the players for each metric. This means that the higher your value is, the better you are in that metric in the Ivy League, and the bigger area your chart has on the plot. Certain metrics where small values considered better (such as losses on own half) are flipped so that bigger area still reflect better value.



4.2 Goalkeepers

Goalkeepers are shown under the **Goalkeeper** tab with two possible visualizations to choose from: scatter plots and radar charts. In the **Goalkeeper** tab all Ivy League goalkeepers are available to analyze, who are above the threshold for minutes played – we set this threshold during our data cleaning process. Goalkeepers have data from all the games they played in an Ivy League team starting the 2021 season.

4.2.1 Scatter Plots

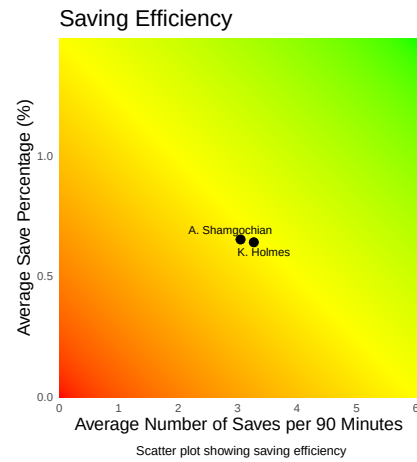
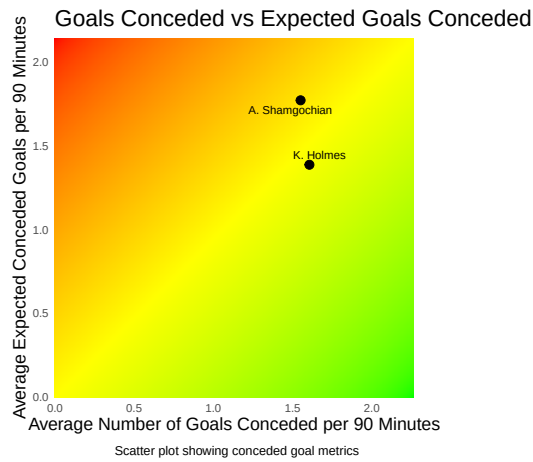
The scatter plot for goalkeepers is very similar to outfield players, only there is slightly less control the user has. The user can still select which goalkeeper(s) to include on the plots, and which variables they want to look at. There are five pre-set plot types for goalkeepers.

Table 3: Preset Goalkeeper Scatter Plot Types and Metrics

Plot Title	X-Axis	Y-Axis
Goals Conceded vs Expected Goals Conceded	goals_conc_per90	xcg_per90
Long Passing Ability	long_per90	long_pct
Short Passing Ability	shortp_per90	shortp_pct
Saving Efficiency	saves_per90	save_pct
Distribution of Goaltending Actions	exits_per90	saves_per90

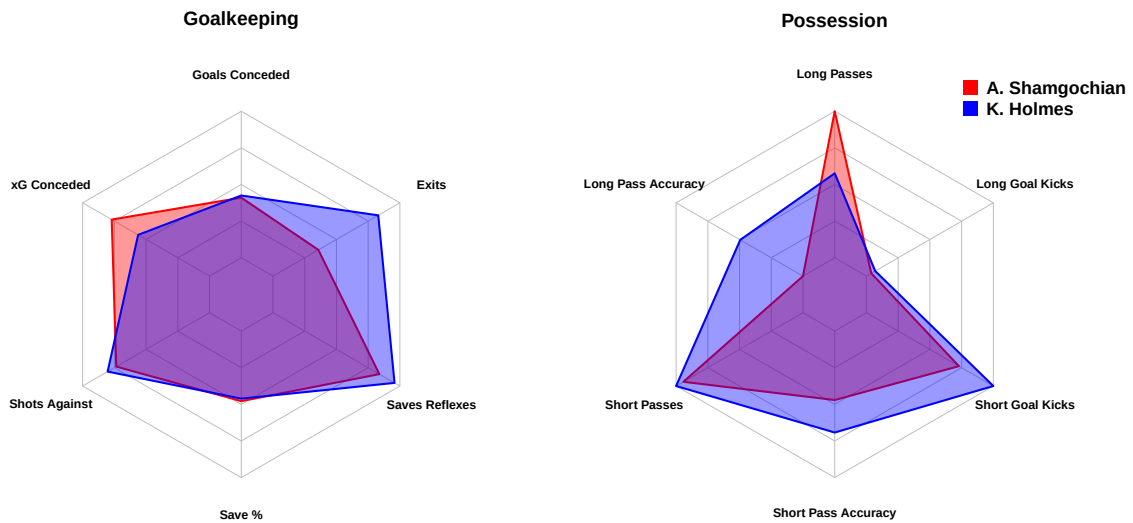
Since all the players on this plot are goalkeepers, it did not make sense to let the user control the position variable. Partially because of this, and because there are a lot less goalkeepers than outfield players, the gradient background is set to represent the league average and cannot

be changed by the user. So a goalkeeper in the green area is good at the specific metric in the Ivy League.



4.2.2 Radar Charts

For goalkeepers, we categorized variables into **Goalkeeping** and **Possession** skills. Apart from this, they work exactly the same as for outfield players. They are percentage based, and where needed the metrics are flipped.



4.3 Teams

Team data can be analyzed under the **Team** tab with two possible methods to choose from: bar charts and radar charts. Here, all Ivy League teams are available for the 2021, 2022, 2023, and 2024 seasons.

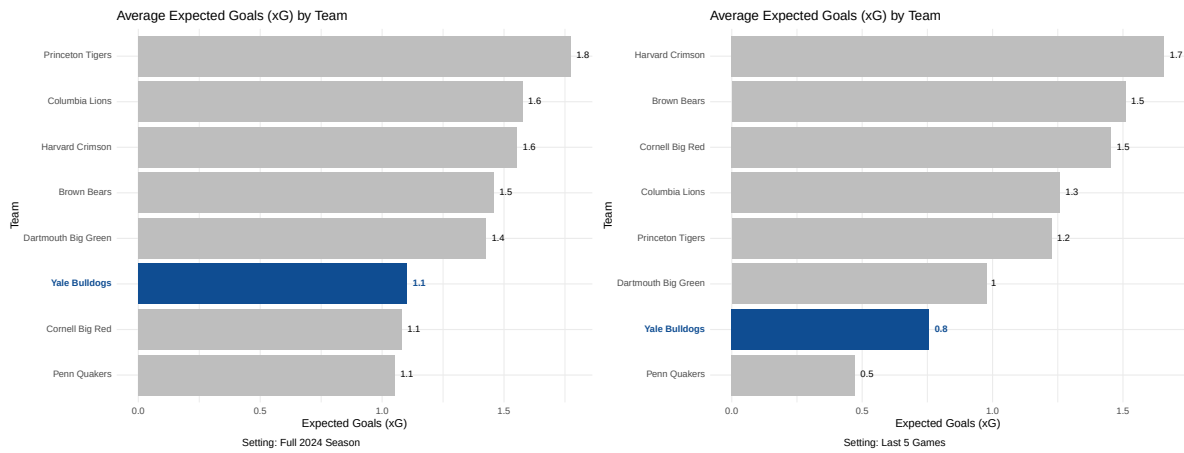
4.3.1 Bar Charts

Bar charts are only available for teams and they all include all eight Ivy League teams. Since our Shiny App is made for Yale coaches, we made sure the bar belonging to Yale is navy blue while all the other ones are gray. There are ten pre-set plot types from which the user can choose which metric they want to look at.

Table 4: Team Bar Chart Metrics and Axis Labels

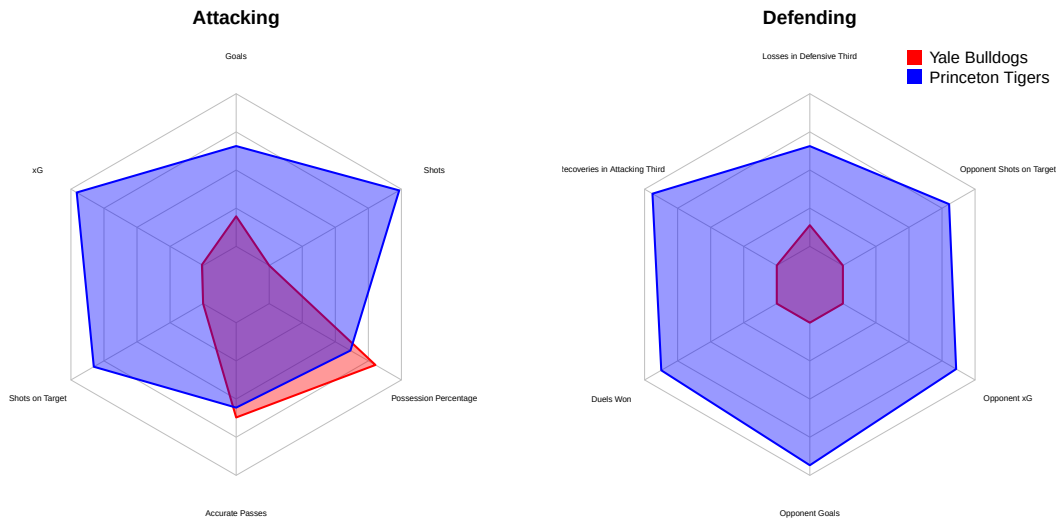
Plot Title	X-Axis
Goals	goals
Expected Goals (xG)	xg
Shots on Target	shots_on_target
Pass Percentage	pass_pct
Possession Percentage	poss_pct
Losses in Low Third	losses_low
Recoveries in High Third	rec_high
Duels Won Percentage	duels_pct
Opponent Goals	opp.goals
Opponent xG	opp.xg
Opponent Shots on Target	opp.shots_on_target

The user can select one or multiple seasons to filter the data for. Alternatively, if the user only wants to look at the **last n** games, they can also do that. We have options to look at the **Last 5**, **Last 10**, or **Last 15** games.



4.3.2 Radar Charts

Radar charts for team work the same as for goalkeepers and outside players. They are percent-age based, the metrics are standardized, and where needed the metrics are flipped. For teams, we decided to put the variables into two categories: **Attacking** and **Defending** attributes. All radar charts in our Shiny App are set up in a way that each team and player will only have one specific polygon associated with them.



4.4 Insights and Interpretation of Visuals

When looking at the right sequence of visualizations, one can learn a lot about teams and individual players. This is one of the biggest reason why we think this platform can help coaches with decision making: looking at non-advanced statistics without having to pay for the more advanced ones still help getting insights that are often overlooked on video. These patterns, clearly visible in the visualizations, help connect raw numbers to tactical realities, offering coaches a more intuitive understanding of where strengths lie and where improvements are needed.

The data shows that Yale is the best team in the league at maintaining possession, consistently recording the highest possession percentages across games. However, Yale also has the second fewest shots on target per game, highlighting a critical weakness: the team is able to keep the ball but struggles to turn possession into chances. Further analysis reveals that defenders dominate in both total passes and overall number of actions per game within the Yale squad. This suggests that Yale's possession-based strategy heavily revolves around circulating the ball within the defensive unit. This ball possession tactic does not work effectively if Yale cannot advance to the middle and attacking third frequently enough. We also see that Yale gives up the second most goals and third most expected goals, which further reinforces that the ball-possession tactic has to improve by possessing the ball higher up the field, or a new tactic should be chosen that would lead to at least conceding less goals.

5 Conclusions, Limitations and Recommendations

The Shiny app developed from this project serves as a useful tool for coaches, helping them make decisions that could directly influence game outcomes. Visuals such as scatter plots and radar charts allow coaches to quickly and intuitively compare players, which is far more effective than analyzing raw Excel spreadsheets.

The main goal of this tool is to assist with roster building and lineup selection throughout the season. However, there are several limitations to consider. The available data is fairly basic, lacking more advanced metrics that could provide deeper tactical insights. Younger players have fewer games recorded, which could skew statistics and may not fully represent their future potential.

The data pipeline is entirely manual: downloading player and team files after every game during the fall season is extremely time-consuming and could introduce delays in updating the visuals. A major potential improvement would be gaining access to the Wyscout API (at a cost of approximately \$5,000 per season). This would allow access not just to more advanced performance metrics, but also to automatic data updates after each game, which would vastly improve efficiency and freshness of insights. The data update is especially important for younger players who do not have a lot of minutes in the original data.

In the future, further improvements to the app could include adding a season filter for scatter plots, adding a position filter for radar charts, and introducing even more user-controlled features to customize analysis based on coaching needs. In the next steps, we would also add a polygon to the radar charts representing the average team/player in the Ivy League. We are very excited to keep on working on this project during next season and see the impact it has on the Yale Women's Soccer team.

6 References

Wyscout. (n.d.). *Football player and team performance data (Excel files)* [Data set]. Wyscout. <https://wyscout.com/>